# DS701 Midterm Team Project

Assigned Wed Oct 7 · due Sun Nov 1, 11:59 pm · orals Fri Oct 16, Fri Oct 23 and Fri Nov 6 in section

**Teams.** You work in a **team of three** (two only by exception), and all members must be in the **same Friday discussion section** — that is where the oral examination happens. Teams self-select: declare your team via the team-declaration form on Gradescope by **Wed Oct 7**. If you cannot find teammates or complete your team, notify course instructors via message on Piazza.

**Grading: 70% team, 30% individual.** The team component (70%) covers the deliverable and your team's presentation of it. The individual component (30%) is your own performance in **three oral checkpoints**, all held in your Friday discussion section. Attendance is mandatory on these three Fridays.

| Checkpoint | Date | Slot | Examines | Weight |
|---|---|---|---|---|
| 1 | **Fri Oct 16** | 5 min | questions, EDA | 5% |
| 2 | **Fri Oct 30** | 5 min | validation plan, baselines, unsupervised | 5% |
| 3 | **Fri Nov 6** | 10 min | results, supervised work, what-ifs | 20% |

At every checkpoint, staff pick which member opens with a short summary, then direct **at least one question to each member by name** (at checkpoint 3, at least two: one about a part you worked on, one about how your approach would change under new conditions). Questions are drawn live and only the named member may answer. Each checkpoint score stands on its own. A member who cannot explain the team's own work receives a low individual score regardless of how good the deliverable is — the orals exist to reward understanding, not polish. Checkpoints 1 and 2 also give your team early feedback while there is still time to act on it.

## The task

Pick **one** dataset from the menu below. Pose **2–3 concrete questions** about it — questions someone might actually pay to have answered, not "apply k-means to X." Then answer them using techniques from Lectures 01–13, and **justify every modeling choice** you make.

Your analysis must include, at minimum:

1. **At least one unsupervised technique** (k-means, hierarchical clustering, GMM/EM, or SVD/PCA/t-SNE used for structure-finding) **and at least one supervised technique** (decision trees/ensembles, k-NN, naive Bayes, SVM, linear or logistic regression with regularization) — each chosen *because it answers one of your questions*, with the choice argued. A tour of every method in scikit-learn is a weaker project than two methods chosen well.
2. **Honest validation.** Split before you fit; never let test data influence any choice (features, hyperparameters, cluster counts). Compare against a sensible **baseline** (predict the mean; majority class; a one-feature rule) before crediting a model. State how sure you are: cross-validation spread, stability of clusters under resampling, sensitivity to a preprocessing choice. A claim without an honesty check is an overclaim.
3. **Stated limitations.** What your evidence does *not* show, and what you would need to show it.

**AI policy.** AI assistance is allowed and encouraged — this project is also practice in directing AI tools productively. But **every AI-assisted artifact (code, analysis, prose) must be understood and justified by your team**: the oral examination checks exactly this, and "the model wrote it" is not an explanation.

## Deliverables

Due **Sun Nov 1, 11:59 pm**. Late submissions accepted up to 48 hours after the deadline with a 10% penalty — but the final oral runs **Fri Nov 6** regardless of when you submit.

**Checkpoint pushes.** Before each of the first two checkpoints, push to your repository by **Thursday 11:59 pm** (late pushes cost the team 2 of its 70 points each):

- **Thu Oct 15**, for checkpoint 1: `QUESTIONS.md` stating your 2–3 questions, each with one sentence on who would act on the answer, and an exploratory analysis notebook that runs top to bottom.
- **Thu Oct 22**, for checkpoint 2: your frozen validation plan as a section of `QUESTIONS.md` (what is held out, and from which commit on you won't touch it), your baseline numbers, and a first pass of your unsupervised analysis.
- **A private GitHub repository**, one per team. Create it as a **private** repository and invite the instructors' GitHub accounts **`trgardos`** and **`Farid-Karimli`** as collaborators so they can review it; do this when you create the repo, because staff read it before checkpoint 1. It contains:
  - **A runnable notebook**: executes top-to-bottom on a fresh kernel with a fixed random seed, including a data-loading cell that downloads the data (or documents exactly where to place it). If we cannot run it, it did not happen.
  - `CONTRIBUTIONS.md`: one section per member stating specifically who did what. The oral draws on it — expect to be asked about the parts you claim.
- **A report, at most 3 pages (PDF)**, structured *question → approach → evidence → limitations* for each of your questions. Figures count toward the limit; do not include an appendix either. We suggest you write the report in markdown in your repo (to faciliate collaboration) and use a tool like Pandoc to convert it to PDF.

- **Submit** the report PDF on Gradescope (one submission per team, all members tagged) with the repository link on the first page.

## Dataset menu — choose one

Four datasets with four different personalities. Harder data handled honestly beats easy data handled superficially — and we grade the honesty, not the difficulty.

### 1. Ames housing — messy tabular prediction

Residential property sales in Ames, Iowa, 2006–2010: **1,460 sales × 81 columns** (sale price plus lot, quality, condition, garage, basement, neighborhood, …). Already in the course repo at `data/ames-housing-data/` (`train.csv`, ~450 KB, with `data_description.txt` defining every column). The mess is the point: missing values that *mean something* (`NA` in `GarageType` means "no garage"), ordinal quality codes pretending to be categories, a skewed target, and far too many columns to use uncritically. L10 used this dataset for a five-minute demo; your job is the real analysis.

*A strong project* treats missingness as information rather than deleting it, and beats a defensible baseline (e.g., median price by neighborhood) before claiming its model works.

### 2. Bank marketing — behavioral prediction with traps

Telemarketing calls from a Portuguese bank's term-deposit campaigns, May 2008–Nov 2010: **41,188 calls × 21 columns** (client demographics, campaign history, economic indicators; target = did the client subscribe). Download: https://archive.ics.uci.edu/static/public/222/bank+marketing.zip (1 MB; use `bank-additional-full.csv` inside). Two traps make this a generalization-discipline exam: only **11.3%** of calls succeed (accuracy is a lie), and the `duration` column is known only *after* the call ends (a model that uses it predicts nothing deployable). The rows are also in time order, spanning a financial crisis.

*A strong project* excludes `duration` with an argument, evaluates with imbalance-appropriate metrics (precision/recall, lift), and checks whether a time-based split tells a different story than a random one.

### 3. Online retail — segmentation from raw transactions

One year of transactions from a UK online gift retailer, Dec 2010–Dec 2011: **~542,000 line items** (invoice, product, quantity, price, customer, country). Download: https://archive.ics.uci.edu/static/public/352/online+retail.zip (23 MB; a single Excel file — `pd.read_excel` needs `openpyxl` and takes a minute or two; save to parquet/CSV once). There is **no feature matrix**: you must build one — aggregate line items to customers (recency, frequency, monetary value are the classic trio), decide what a cancellation (negative quantity) means, and face the ~25% of rows with no customer ID.

*A strong project* justifies its feature construction as carefully as its model, picks the number of clusters with more than an elbow, and builds any predictive label strictly across a time boundary.

**4. Smartphone activity recognition — high dimensions, honest splits**

Smartphone accelerometer/gyroscope recordings of **30 subjects** doing six activities (walking, stairs up/down, sitting, standing, lying): **10,299 windows × 561 engineered features**, with subject IDs and a provided subject-disjoint train/test split. Download: https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip (58 MB; ignore the `Inertial Signals/` folders — the raw waveforms — and work with `X_train`/`X_test`, ~70 MB as text). This is the dimensionality-reduction playground: 561 correlated features begging for SVD/PCA/t-SNE. And it hides the sharpest validation lesson on the menu: windows from the same person are highly similar, so a random row split leaks *people* across train and test.

*A strong project* reports both splits and explains the gap as a claim about generalizing to new people, rather than quietly reporting the flattering number.

## How it is graded

| Dimension | Scored | What full marks sound like |
|---|---|---|
| Technique appropriateness | team | methods fit the questions; alternatives were considered and rejected for reasons |
| Justification quality | team | choices argued from the data and diagnostics — including the AI-assisted parts |
| Result integrity | team | every claim matches evidence in the notebook; limitations stated plainly |
| Individual comprehension | **each member** | you can explain your part *and* the team's core method, and reason about a variation |

Team deliverable + presentation (the three team dimensions) → **70%**; your own individual comprehension across the three oral checkpoints → **30%** (5% + 5% + 20%). Everyone on the team owns the headline claim.

## Suggested timeline

- **Week 1 (Oct 7–15).** Declare your team (by Oct 7). Set up the private repo and invite `trgardos` and `Farid-Karimli`. Pick the dataset, read its documentation, do exploratory analysis, and write down your 2–3 questions *before* modeling. Push `QUESTIONS.md` and the EDA notebook by **Thu Oct 15**.
- **Fri Oct 16 — checkpoint 1** in section: questions and EDA.
- **Week 2 (Oct 16–22).** Freeze your validation plan (what is held out, and from what point on you won't touch it). Establish baselines. Run your unsupervised analysis. Push by **Thu Oct 22**.
- **Fri Oct 23 — checkpoint 2** in section: validation plan, baselines, unsupervised first pass.
- **Week 3 (Oct 23–27).** Supervised modeling. Note that regularization (Oct 21), SVD (Oct 26) and PCA/t-SNE (Oct 28) arrive in class mid-project. The SVD and PCA/t-SNE recordings are posted ahead of their sessions, so watch them early or plan around them.

- **Week 4 (Oct 28–Nov 1).** Honesty checks, limitations, the 3-page report. Restart the kernel and run the notebook top-to-bottom. Submit — quiz 2 is Mon Nov 2, so do not plan to use the late window.
- **Fri Nov 6 — checkpoint 3**, the final oral. Prepare for it: every member should be able to walk through the *entire* notebook, not just their own cells.